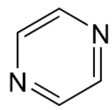
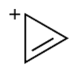
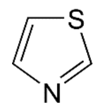
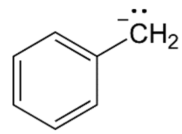
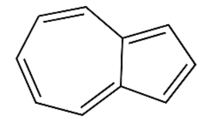
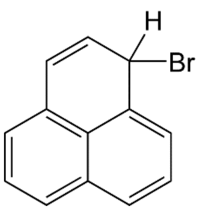
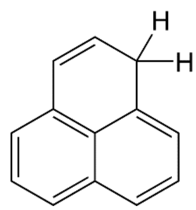
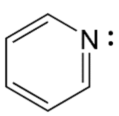
1. Classify the following as aromatic, anti-aromatic or non-aromatic.

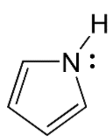






2. What ions could be formed from the following? Predict which would more easily form and explain why.

3. Pyridine is a stronger base than pyrrole. Give an explanation invoking Hückel's rule and aromaticity.


pyridine


pyrrole

3. Show the mechanism to produce the electrophile in each of the following.

a) HNO_3, H_2SO_4
b) Br_2, $FeBr_3$
c) 1-chloro-2,3-dimethylpentane, $AlCl_3$

4. Draw the mechanism for the following electrophilic aromatic substitutions.